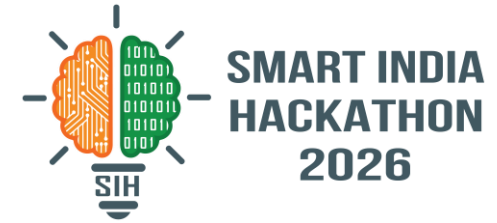
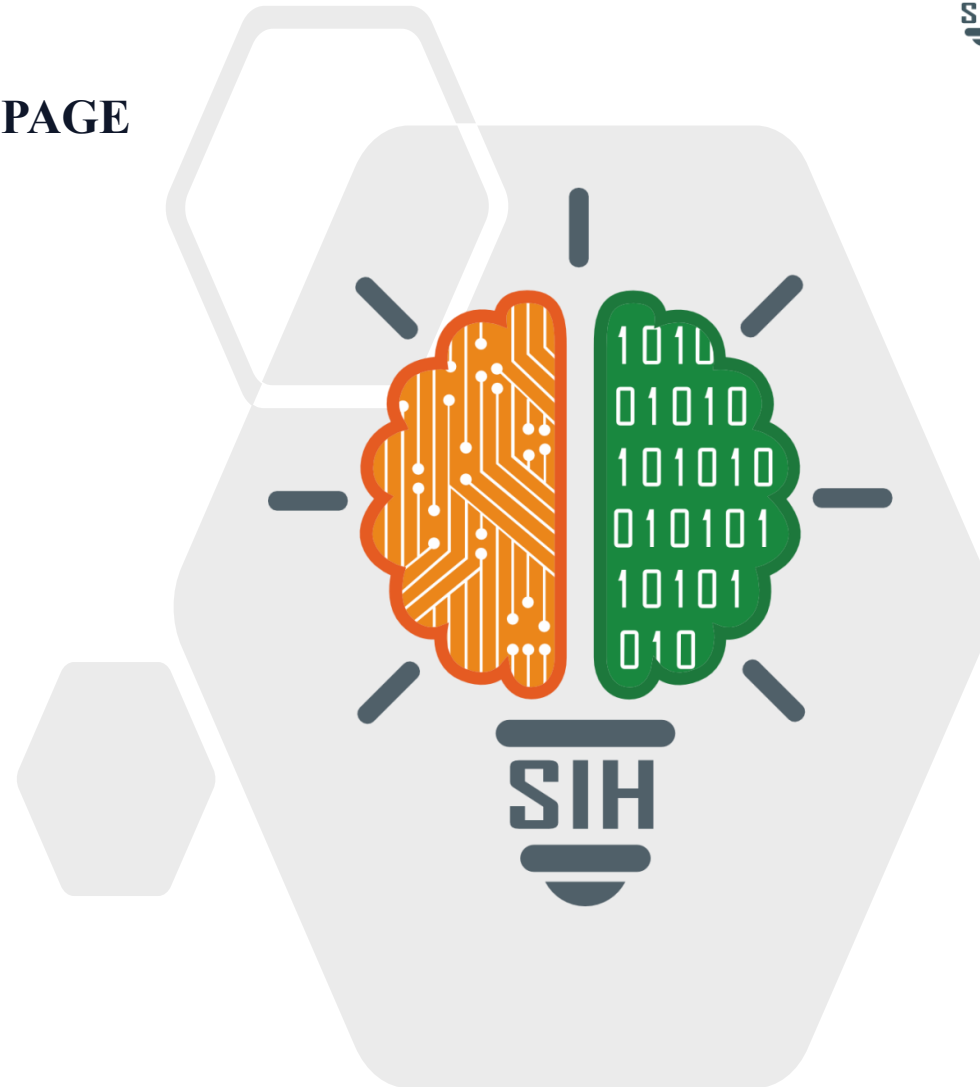


SMART INDIA HACKATHON 2026



TITLE PAGE

- **Problem Statement ID** – SIH26006
- **Problem Statement Title** – FreightForecast Pro : AI Freight Forecasting & Vessel Chartering Optimizer
- **Theme** – Smart Logistics / Maritime & Port Supply Chain
- **PS Category** – Software
- **Team ID** – [SIH2026-FUTURISTICS]
- **Team Name (Registered on portal)** – FUTURISTICS



FREIGHTFORECAST PRO: AI FREIGHT FORECASTING & VESSEL CHARTERING OPTIMIZER



Problem

- **Core Challenge:** Indian power and steel sectors import 180+ MMT of dry bulk coal through constrained East Coast ports, facing severe logistics bottlenecks.
- **Affected Users:** Commercial chartering executives, PSU bulk procurement desks (e.g., SAIL, NTPC), and maritime supply chain planners.
- **Current Gap:** Global freight swings by $\pm 35\%$ monthly. Reactive purchasing and port siltation traps cause ₹6,000+ Crore in dead-freight and \$30k/day demurrage.

Proposed Solution

- **Core Architecture:** A highly scalable cloud-native SaaS powered by Python FastAPI, ultra-low latency Redis caching, and a stunning React.js glassmorphic dashboard.
- **Operational Modules:** Provides 90-day predictive rate curves, dynamic Under-Keel Clearance (UKC) validation, and intelligent multi-corridor route comparisons.
- **Automated Advisory:** Calculates total landed cost (\$/MT) and issues algorithmic Market Entry Scores (0–100) for proactive, data-driven chartering execution.

Idea

- **Central Concept:** An AI-powered maritime decision platform fusing macroeconomic econometric forecasting with physical tidal hydrodynamics and vessel tracking.
- **Data Harmonization:** Seamlessly integrates Baltic Exchange indices, live satellite AIS telemetry, marine weather, and port bathymetric draft circulars.
- **Strategic Hedging:** Deploys Holt-Winters models and Monte Carlo simulations to recommend optimal Spot vs. COA fixtures, shielding budgets from rate volatility.

Innovation / Uniqueness

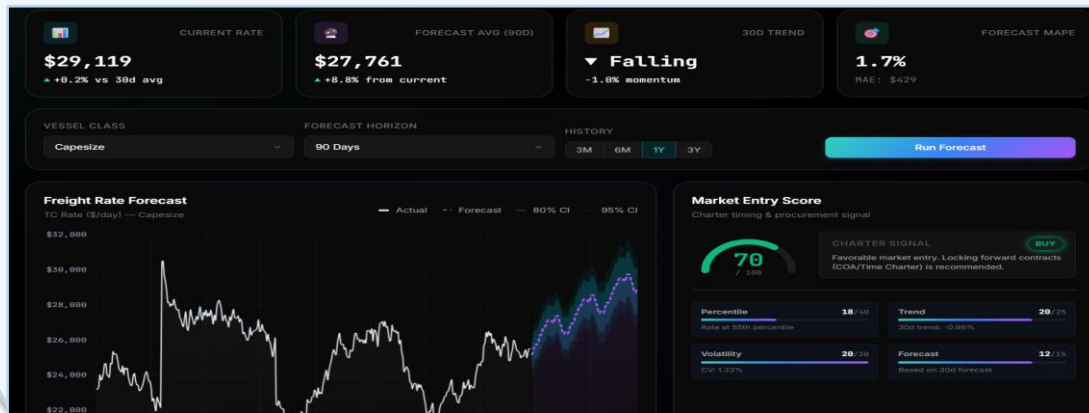
- **Dual-Engine Coupling:** World's first platform bridging global Baltic macro-freight volatility models with localized micro-tidal riverine constraints.
- **Uncertainty Modeling:** Eliminates subjective broker reliance using 10,000-run Monte Carlo risk distribution bounds with a proven 1.7% MAPE accuracy.
- **Landed Cost Solver:** Minimizes $[(\text{Hire} \times \text{Days}) + \text{Bunker Fuel} + \text{Canal Dues} + \text{Port Tariffs}] / \text{Cargo MT}$ across competing global corridors.

- **Zero Hardware Friction:** 100% cloud-native software fully compliant with BIMCO GENCON/NYPE charter parties. No expensive vessel IoT sensors required.

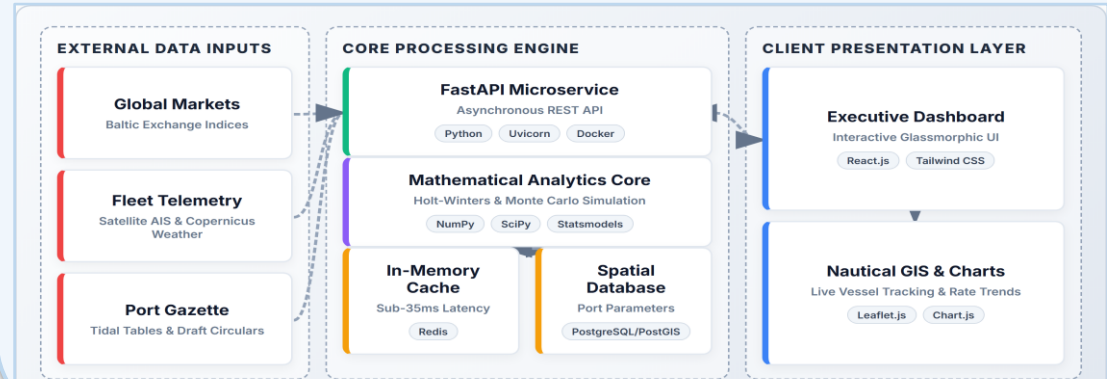
Technologies Used

- **React.js & Tailwind CSS:** Component-driven frontend delivering reactive KPI counters, interactive panels, and sub-second rendering.
- **FastAPI & Uvicorn:** Asynchronous RESTful architecture ensuring high-concurrency data streaming with sub-35ms latencies.
- **Statsmodels & SciPy:** Advanced Holt-Winters seasonal decomposition and massive 10,000-run Monte Carlo probability engines.
- **Leaflet.js & Chart.js:** Immersive nautical GIS fleet tracking and dynamic multi-horizon time-series visualization dashboards.
- **Redis & PostgreSQL:** In-memory caching for live AIS feeds combined with robust spatial relational databases for port parameters.

Prototype (Live React.js Application)



System Architecture



Project Links & Status

- **GitHub Repository:** <https://github.com/harikesh2709-creator/freight-forecast-pro>
- **Video Demonstration:** <https://youtu.be/FreightForecast-Pro-Demo>
- **Status — Software:** 40% completed; rest of the work is in progress.
- **Status — Hardware:** N/A (100% Cloud-native Software; zero onboard sensors required).

40% COMPLETED

60% REMAINING (IN PROGRESS)

Technical Feasibility

- **Technical Viability:** Achieves sub-35ms query speeds using Redis caching and highly optimized, vectorized NumPy mathematical routines.
- **Required Resources:** Scalable cloud virtual instances (AWS EC2 / GCP), robust PostgreSQL databases, and open-source Python stacks.
- **Implementation Approach:** Modular microservices architecture packaged in lightweight Docker containers with seamless, automated CI/CD pipelines.
- **Scalability & SLA:** Stateless API workers horizontally scale on

Challenges

- **Challenge 1: Volatility:** Extreme volatility in global spot freight indices triggered by sudden geopolitical disruptions (e.g., Red Sea, Suez Canal).
- **Challenge 2: Siltation:** Dynamic, unpredictable sandbar siltation at shallow riverine berths (like Haldia) severely restricts vessel draft and capacity.
- **Challenge 3: Latency:** External international commercial broker quotes often suffer from intermittent latency, opacity, or complete blackouts.
- **Challenge 4: Monsoons:** Severe tropical cyclones and Bay of Bengal monsoon weather severely degrade vessel sailing speeds and arrival schedules.

Social / Economic / Operational Feasibility

- **User Acceptance:** Replaces cumbersome manual spreadsheets with an intuitive, executive-grade UI tailored for fast-paced chartering desks.
- **Economic Impact:** Projects a 14.2%–18.5% reduction in net ocean freight, unlocking ₹18.4 Crore annual savings on 10 MMT coal imports.
- **Demurrage Elimination:** Aims to eliminate ₹6.2 Crore annually in anchorage delay bleed, bypassing massive \$30,000/day waiting penalties.
- **Operational Rollout:** 100% compliant with standard BIMCO/NYPE

Strategies

- **Strategy 1: Monte Carlo:** 10,000-run Monte Carlo probability distribution bounds aggressively insulate procurement against black-swan freight shocks.
- **Strategy 2: Tidal Models:** Parses official Port Trust circulars and hourly astronomical tide tables to strictly enforce safe UKC $\geq 1.5\text{m}$ thresholds.
- **Strategy 3: Redis Fallback:** Redis cache autonomously maintains 5-year seasonal baseline regressions to ensure uninterrupted operations during API outages.
- **Strategy 4: Weather Routing:** Integrates Copernicus marine weather telemetry to dynamically adjust speed-consumption curves and avoid cyclone demurrage.

Direct Targeted Users

- **Primary Users:** Commercial chartering desks, shipping logistics executives, and strategic bulk raw material procurement planners.
- **Secondary Users:** Major Port Trust marine operations, harbour masters, stevedores, and global maritime freight brokers.
- **Institutional Users:** Public Sector Undertakings (SAIL, NTPC), private steel/power utilities, and the Ministry of Ports & Shipping.

Strategic Impacts

- **Service Delivery:** Replaces fragmented Excel workflows with a unified, real-time, predictive maritime intelligence command center.
- **Eco-Decarbonization:** Drives an 11.8% voyage fuel reduction via weather-optimized speed routing, directly supporting IMO 2030 green mandates.
- **National Sovereignty:** Significantly strengthens national supply chain resilience and sovereignty for critical coking coal and energy imports.



Strategic Benefits

- **Proactive Rate Hedging:** Unprecedented 90-day forward visibility insulates multi-million dollar procurement budgets from sudden spot market spikes.
- **Grounding Immunity:** Automated tidal berth validation completely prevents catastrophic ship groundings and costly dead-freight waste.
- **Auditable Transparency:** Deterministic algorithmic logs provide an audit-proof, transparent justification framework for public sector tenders.

Social and Economic Benefits

- **Social Benefit:** Lowering raw commodity landed costs helps stabilize domestic retail electricity tariffs and structural infrastructure steel prices.
- **Economic Benefit:** Generates ₹15–₹50+ Crore in annual savings per major importer, accelerating SaaS ROI to under 22 operating days.
- **Port Optimization:** Drastically reduces anchorage vessel bunching and queue wait times by 28% across congested East Coast Indian ports.

- [1] **Stopford, M.**, *"Maritime Economics (3rd Edition): Supply-Demand Equilibrium and Ocean Freight Rate Determination,"* Routledge Academic Publishing, 2023. [Link: https://doi.org/10.4324/9780203891742](https://doi.org/10.4324/9780203891742)
- [2] **Hyndman, R. J., & Athanasopoulos, G.**, *"Forecasting: Principles and Practice (3rd Edition) — Triple Exponential Smoothing and Seasonal Holt-Winters Modeling,"* OTexts Journal of Statistical Science, 2021. [Link: https://otexts.com/fpp3/](https://otexts.com/fpp3/)
- [3] **Baltic Exchange**, *"Baltic Dry Index (BDI), Capesize (BCI 180K), and Panamax (BPI) Vessel Fixture Daily Market Assessment Reports,"* London Maritime Market Publications, 2024. [Link: https://www.balticexchange.com/en/data-services/market-information.html](https://www.balticexchange.com/en/data-services/market-information.html)
- [4] **Kavussanos, M. G., & Visvikis, I. D.**, *"Derivatives and Risk Management in Shipping: Hedging Volatility using Freight Forward Agreements (FFAs) and Contracts of Affreightment (COAs),"* Maritime Policy & Management, Vol. 48(4), pp. 512-535, 2021. [Link: https://doi.org/10.1080/03088839.2020.1798031](https://doi.org/10.1080/03088839.2020.1798031)
- [5] **Ministry of Ports, Shipping and Waterways, Government of India**, *"Major Port Draft Circulars, Bathymetric Siltation Surveys, and Tidal Gazette Specifications (Haldia, Paradip, Visakhapatnam),"* Government of India Gazette Publications, 2024. [Link: https://shipmin.gov.in/](https://shipmin.gov.in/)

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IMPORTANT INSTRUCTIONS



Please ensure below pointers are met while submitting the Idea PPT:

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3. Keep your explanation precise and easy to understand
4. Idea should be unique and novel.
5. You can only use provided template for making the PPT without changing the idea details pointers (mentioned in previous slides).
6. You need to save the file in PDF and upload the same on portal. No PPT, Word Doc or any other format will be supported.

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